Existing foundation stability

15 m base-hinged high mast (Abacus type), 2 floodlights, on the base of Section K-K

Prepared by	DIMO Lighting Solutions – Design Tools (Poles & high masts, Existing foundation)
Date	20 October 2026
Status	Indicative assessment for the design team – to be checked and signed by a chartered structural engineer
Basis	BS EN 1991-1-4 wind · ILP PLG 07 hinge case · BS EN 1993-1-8 · BS EN 1992-1-1 / -4 · BS EN 1997-1

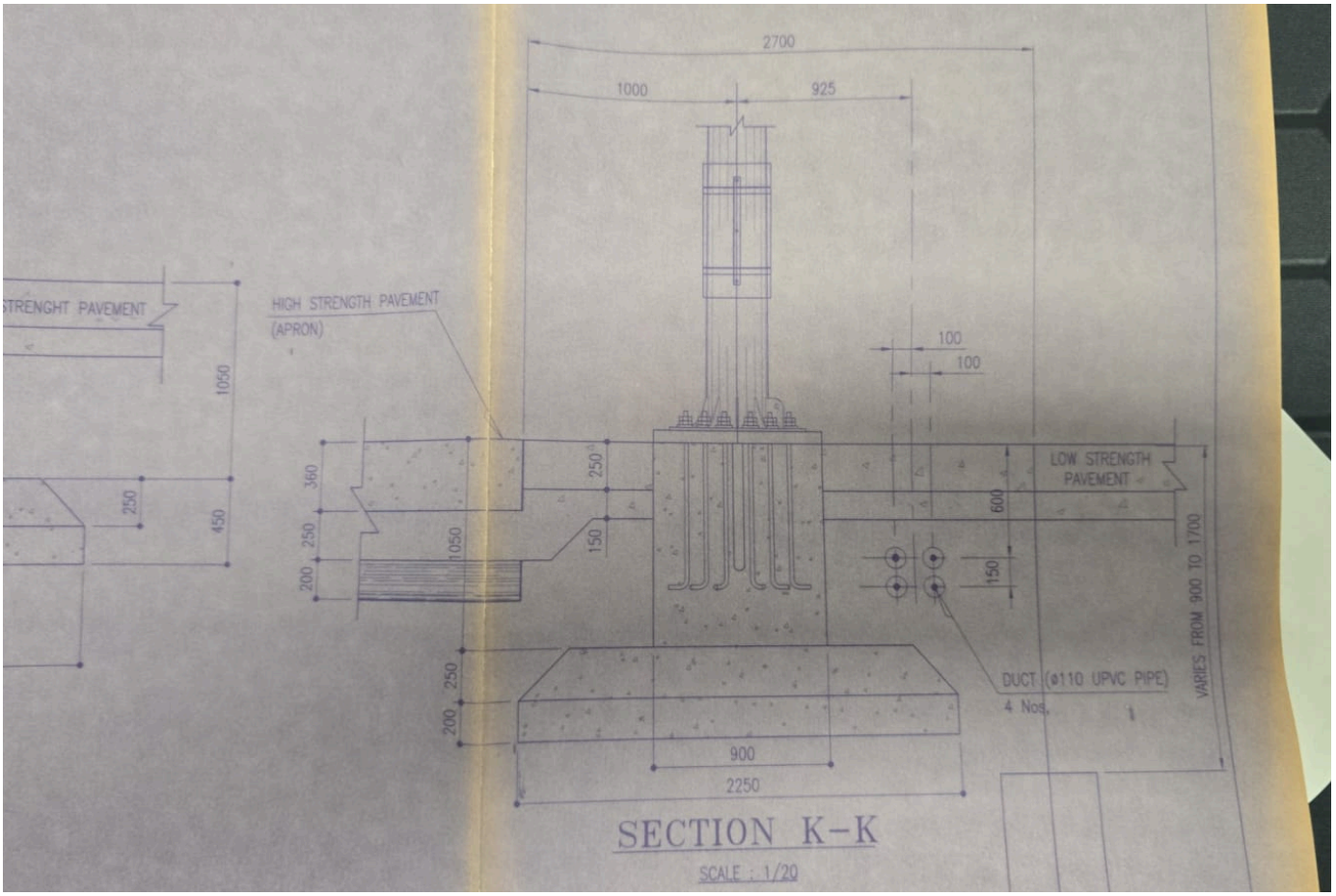
Conclusion: the existing base is ADEQUATE for a 15 m base-hinged mast with 2 × 12 kg floodlights (0.30 m² each).

In the Colombo wind zone (38 m/s gust) every check passes with good margin: overturning ratio **2.49** (needs ≥ 1.0) – **1.64** even at the shallow 0.9 m side; soil pressure **59 kPa** against 150 kPa allowable; holding-down bolts **14 %** used. The lowering (hinge) case, 32.4 kNm, is smaller than the wind case, 39.5 kNm.

It also passes in wind zone 2 (43 m/s). Only in the cyclone zone 1 (49 m/s) **and** at the shallow 0.9 m depth does it fall just short (overturning 0.96) – not the case for Colombo. **Conditions:** the new hinge base must be fitted to the existing bolts (see §5), and the existing concrete and bolts confirmed sound.

1. The existing base – as read from the drawing

Item	Read from Section K-K (scale 1:20, photo)	Used in the check
Pad (footing)	2250 mm wide; 200 mm at the edge, sloping up 250 mm to about 450 mm at the pedestal	2.25 × 2.25 m (assumed square), 0.45 m thick
Pedestal	900 mm square, about 1.1 m high, cast-in hooked (J) bolts	0.90 m square, top 75 mm above the apron
Depth	“Varies from 900 to 1700” below the pavement	1.6 m to the underside; 0.9 m checked as the shallow side
Holding-down bolts	Number, size and PCD not legible on the photo; embedment ≈ 850 mm	8 × M30 grade 8.8 on 620 mm PCD, hef 0.85 m (assumed)
Concrete / reinforcement	Not legible	C25/30; bottom H16 @ 200 each way (assumed)
Surroundings	250 mm high-strength apron, low-strength pavement, 4 Ø110 uPVC ducts	Not counted (conservative)
Soil	Not given	Granular, φ′ 30°, γ 18 kN/m³, allowable bearing 150 kPa



The drawing as received (Section K-K), turned upright.

2. The mast and its loads

Item	Value
Mast	15 m base-hinged, octagonal S355 shaft 350 → 150 mm, ≈ 477 kg; lowered by a counterbalance ram (about 80 mm bore at 110 bar)
Head load	2 floodlights × 12 kg, 0.30 m² each (Cf 1.0) + head frame 20 kg, 0.10 m² (assumed) = **44 kg, 0.70 m²**
Abacus range	**HL250** at 15 m: 100 kg and 0.718 m² at 45 m/s → 1.01 m² at the site gust – carries this head; counterbalance RLH5; base bolts on a 400 mm PCD
Wind	Sri Lanka zone 3 (incl. Colombo) – 38 m/s 3-s gust → vb,0 26.6 m/s; 50-year return; qp at 15 m ≈ 0.87 kPa
Base actions – wind case (unfactored)	Moment **39.5 kNm**, shear 4.7 kN, vertical 5.1 kN · design moment MEd 55.3 kNm
Base actions – lowering (hinge) case	32.4 kNm unfactored (51.6 kNm with dead × 1.6) – less than the wind case, so wind governs
Mast top deflection / frequency	249 mm under the design wind · f1 ≈ 1.36 Hz

3. Stability checks of the existing base – 15 m mast, Colombo wind

Check	Calculation	Result
Existing bolts 8 × M30 grade 8.8 on PCD 620 mm – tension EN 1993-1-8 Table 3.4	Ft,Ed 44.0 kN ≤ Ft,Rd 323.1 kN (0.14)	✓ Pass
Existing bolts – shear + tension EN 1993-1-8 6.2.2(7)	0.11	✓ Pass
Concrete cone – embedment 850 mm EN 1992-4 §7.2.1.4	one bolt 44.0 kN ≤ NRd,c 735.2 kN (cracked, γMc 1.5; spacing ≥ 3 hef assumed – else check the group)	✓ Pass
Bolt circle Fit	the new hinge base needs PCD 510 mm with 8 × M30; the existing is 620 mm with 8 × M30 – drill the new plate to the existing bolts (re-check the plate) or fit an adaptor plate	! Note
Overturning – pad 2.25 × 2.25 × 0.45 m at 1.6 m EN 1997-1 §2.4.7.2 EQU	0.9 G × B/2 ≥ 1.5 M: ratio 2.49 (G 175 kN with soil over the pad)	✓ Pass
Bearing EN 1997-1 §6.5.2	qmax 59 kPa ≤ 150 kPa; e 0.27 m ≤ B/4 0.56 m	✓ Pass
Sliding EN 1997-1 §6.5.3	1.5 Hk 7.0 kN ≤ 52.1 kN (tan(⅔ φ') 0.36)	✓ Pass
Pad bending – bottom H16 @ 200 EN 1992-1-1 §6.1	MEd 19 ≤ MRd 144 kNm/m (d 367 mm, As 1005 mm²/m)	✓ Pass
Pad one-way shear EN 1992-1-1 §6.2.2	vEd 0.070 ≤ vRd,c 0.401 MPa	✓ Pass
Condition of the existing base BS EN 12504-1 · BS EN 13791	core the concrete for its strength, expose the bolts for corrosion and thread, check level and cracks before re-use – the values above assume sound concrete of the grade entered	! Note

Overturning: 0.9 × (concrete + soil over the pad + mast weight) × B/2 ≥ 1.5 × (moment + shear × depth) – BS EN 1997-1 EQU. Bearing: pressure under the pad with the load eccentricity, at least 75 % of the pad in contact. Bolts: BS EN 1993-1-8 with the anchor-bolt shear factor; concrete cone BS EN 1992-4 (cracked concrete, γMc 1.5).

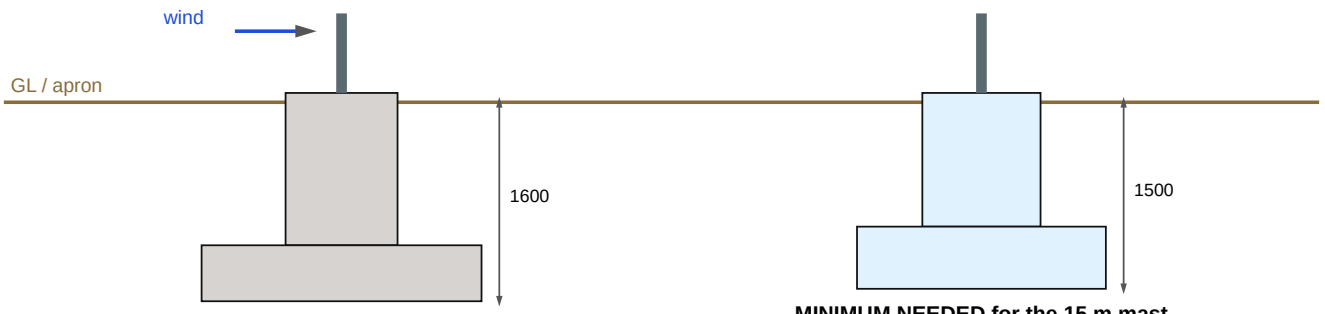
4. Sensitivity – depth and wind zone

Case	Wind	Depth m	Base moment kNm	Overturning (≥ 1.0)	qmax kPa (≤ 150)	Bolts used	Verdict
15 m – 2 × 12 kg / 0.30 m² (this mast)	Zone 3 (38 m/s)	1.6	39.5	2.49	59	0.14	Adequate

15 m – shallow side of the base	Zone 3 (38 m/s)	0.9	39.5	1.64	44	0.14	**Adequate**
15 m – 2 × 12 kg / 0.30 m²	Zone 2 (43 m/s)	1.6	51.9	1.90	67	0.18	**Adequate**
15 m – 2 × 12 kg / 0.30 m²	Zone 1 (49 m/s, cyclone)	1.6	67.4	1.46	78	0.23	**Adequate**
15 m – shallow side	Zone 1 (49 m/s, cyclone)	0.9	67.4	0.96	75	0.23	**Not adequate – Overturning, Bearing**
18 m HL330 – 2 × ALO 3 + 2 × ALO 1 (BIA calc D4980C5)	Zone 3 (38 m/s)	1.6	62.9	1.63	74	0.22	**Adequate**

For reference, Abacus calculation D4980C5 (18 m HL330, SRI LANKA BIA) gives 62 kNm at the base; this method gives 63 kNm for the same mast.

5. Fitting the mast to the base



EXISTING (Section K-K)
2.25 × 2.25 × 0.45 m pad at 1.6 m · pedestal 0.9 m

MINIMUM NEEDED for the 15 m mast
2 × 2 × 0.5 m pad at 1.5 m · pedestal 0.95 m

Item	What to do
Bolt pattern	The HL250 hinge base uses a 400 mm PCD; the existing bolts (not legible – assumed 8 × M30 on 620 mm) will differ. Survey the bolts, then either have Abacus drill the base plate to the existing pattern (they re-check the plate) or fit a galvanized adaptor plate designed for the base actions above.
Bolt condition	Expose, clean and inspect; replace nuts and washers; check thread length for double nuts and levelling; reject bolts with section loss.
Grout	Non-shrink grout ≤ 50 mm under the base plate after levelling; drain path kept.
Lowering zone	About 17 m clear in the lowering direction with a rest for the head; the apron is suitable to work on; no lowering in wind of 30 km/h or more.
Ducts and earthing	The four Ø110 ducts enter beside the pedestal – keep them clear of the ram base; earth the mast to the site earth (≤ 10 Ω).

6. To confirm before installation

- The pad plan shape (square or round) and the depth at each mast position (900–1700 mm on the drawing).
- The existing holding-down bolts: number, size, grade, PCD, projection and condition.
- Concrete strength (cores to BS EN 12504-1) and the pad reinforcement (drawing or cover-meter survey); cracks or spalling around the pedestal.
- Soil bearing value from a site investigation (150 kPa assumed; the base needs only about 59 kPa).
- The floodlights' tested windage at their aiming angle and the head frame weight / windage from Abacus (20 kg, 0.10 m² assumed).

Limitations. Indicative assessment made with DIMO Design Tools from a photograph of the drawing; dimensions not legible on the photo were assumed as stated. Abacus's own calculation for the chosen mast, head and site governs the mast and its bolts; the re-use of the foundation is to be checked and signed by a chartered structural engineer.